



Bertrand Model



Outlines

- Introduction
- Definition
- Assumptions



Bertrand model

- The Bertrand model, named after Joseph Louis François Bertrand, is a theory in economics used to analyze competition between firms in an oligopoly market. In particular, it focuses on the pricing strategies of firms in industries where there are only a few dominant players.



Bertrand model

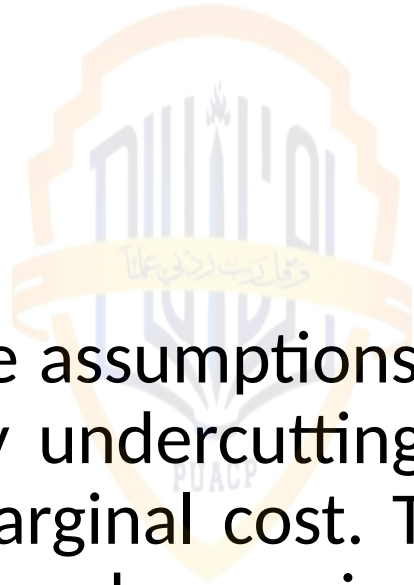
- The Bertrand model contrasts with the more commonly known Cournot model, which assumes that firms compete by choosing quantities of goods to produce rather than prices. In the Bertrand model, firms simultaneously and independently set prices for their products, assuming that consumers will always choose the product with the lowest price.



Key Assumptions

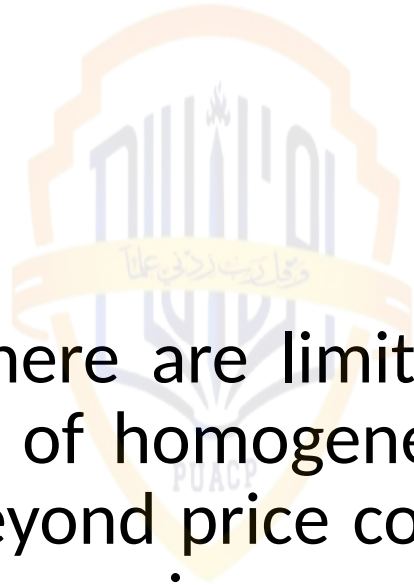
- Key assumptions of the Bertrand model include:
- Homogeneous products: Firms produce identical or perfectly substitutable goods.
- Simultaneous pricing decisions: Firms set prices simultaneously, and consumers make purchasing decisions based solely on the price of the product.
- No capacity constraints: Firms can supply any quantity of the product at the chosen price.
- Rational behavior: Firms aim to maximize their profits.





- Under these assumptions, the Bertrand model predicts that firms will compete by undercutting each other's prices until prices are driven down to marginal cost. This leads to a situation known as Bertrand competition, where prices are driven down to the level of marginal cost, resulting in zero economic profit for firms in the long run.





- However, there are limitations to the Bertrand model, such as the assumption of homogeneous products and the absence of strategic behavior beyond price competition. In real-world markets, firms may engage in non-price competition, product differentiation, or strategic pricing to gain a competitive advantage.

